

EXP 1

Objective 1

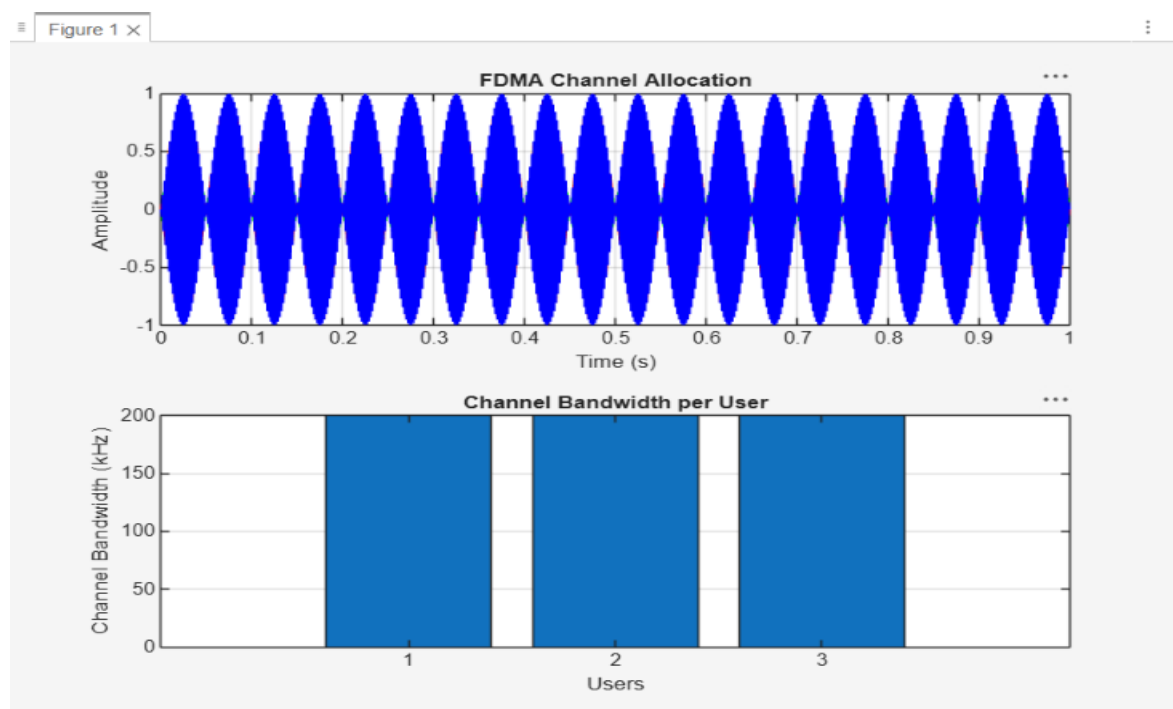
Program

```
clc; clear; close all;
fs=1000; t=0:1/fs:1;
TBW=600; Users=3;

CBW=TBW/Users;
fc=[100 300 500];
m=sin(2*pi*10*t);
s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')
subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)),grid on

xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')
```

OUTPUT



Objective 2

Program

```
clc; clear; close all;  
BW = 200;  
Users = 3;  
M = 2;  
DataRate = BW*log2(M);  
Rate = DataRate*ones(1,Users);  
disp(' User  Data Rate  
(kbps)') disp([(1:Users)'  
Rate'])  
figure  
bar(1:Users,Rat  
e)  
xlabel('Users')  
ylabel('Data Rate (kbps)')  
title('FDMA Data Rate for Each  
User') grid on
```

OUTPUT

